

Changes in prefrontal cortical behaviour depend upon familiarity on a bimanual co-ordination task: An fNIRS study[☆]

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To investigate neurocognitive mechanisms associated with task-related expertise development, this paper investigates serial changes in prefrontal activation patterns using functional near infrared spectroscopy (fNIRS). We evaluate cortical function in 62 healthy subjects with varying experience during serial evaluations of a knot-tying task. All tasks were performed bimanually and self paced, with fixed episodes of motor rest for five repetitions. Improvements in technical skill were evaluated using dexterity indices to quantify time, total movements and pathlength required to complete trials. Significant improvements in technical skills were observed in novices between the 2nd and 3rd trials, associated with increasing task familiarity. In trained subjects, minimal fluctuation in task-related oxyhaemoglobin (HbO₂) and deoxyhaemoglobin (HHb) changes were observed in association with more stable task performance. In contrast, two significant transitions in prefrontal haemodynamic change were observed in novices. Greater task-related increases in HbO₂ and decreases in HHb were identified on the second trial compared to the first. Relative decreases in HbO₂ and increases in HHb change were observed between the third and fourth, and fourth and fifth trials respectively. These data suggest that prefrontal processing across five

knot-tying trials is influenced by the level of experience on a task. Modifications in prefrontal activation appear to confer technical performance adaptation in novices.

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Introduction

Motor learning is a set of internal processes that leads to a relatively permanent change in the learner's capacity for skilled performance (Schmidt and Lee, 2005). Brain activation foci are dynamic and change according to the phase or stage of motor skills learning (Halsband and Lange, 2006). The initial phases of motor skills learning are associated with widespread activations in cortical regions known to be associated with attention and control, such as the prefrontal cortex (PFC) and anterior cingulate cortex (ACC) (Kelly and Garavan, 2005; Halsband and Lange, 2006). These regions support novel cognitive demands in the initial phases of skills learning, during which the task is intellectualised and performance is erratic (Kelly and Garavan, 2005; Halsband and Lange, 2006). Learning complex motor skills necessitates attention to actions (Rowe et al., 2002), visuomotor integration, visuospatial working memory (VSWM) (Baddeley, 1986), correct response selection, performance monitoring, adaptive decision making and behavioural modification (Walton et al., 2007), all of which rely upon the PFC and ACC. Attenuation in these regions accompanies automated performance and a shift in cortical processing to a less effortful, more passive mode of attention (Kelly and Garavan, 2005; Halsband and Lange, 2006).

Studies of motor sequence learning have investigated activation foci associated with sequential movements of the hands and fingers in a lying position, using either positron emission tomography (PET) (Jenkins et al., 1994; Grafton et al., 2002) or functional magnetic resonance (fMRI) imaging (Karni et al., 1995; Sun et al., 2007). These experiments, however, are limited by the confines of the scanner and generally not suitable for motor learning tasks that

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require postural control (Hatakenaka et al., 2007). Functional Near Infrared Spectroscopy (fNIRS) enables assessment of cortical function in ambulant subjects, allowing motor tasks that require complex dynamic movements to be evaluated in real time (Strangman et al., 2002). In comparison to other neuroimaging techniques fNIRS is portable, discreet, tolerant to motion artefact and permits serial assessments of tasks in relaxed and realistic environments.

Paradigms investigated using fNIRS have comprised simple motor tasks such as self paced (Watanabe et al., 1996) or externally paced finger tapping (Toronov et al., 2001), finger to thumb opposition (Obrig et al., 1996) and apple-peeling (Okamoto et al., 2004a,b), all of which require minimal cognitive input. These studies were aimed at investigating the typical fNIRS response to brain activation and were not designed to evaluate learning-related neuroplasticity. Recently, Hatakenaka et al. (2007) demonstrated the potential of fNIRS to investigate cortical plasticity associated with learning a pursuit rotor (PR) task. Modification in activation foci from the pre-supplementary area (pre-SMA) to the supplementary area (SMA) was observed in association with improved PR performance (Hatakenaka et al., 2007). Interestingly, minimal change was observed in PFC activation. Unlike PR, which is learnt implicitly through trial and error, technical maneuvers in surgery are learnt explicitly through observation and imitation, followed by refinements made through further practice.

Elucidating the neurocognitive mechanisms associated with expertise development in surgery may have implications for assessment and selection of future trainees. However, the cortical correlates of technical skills learning in surgery have yet to be delineated since evaluating cortical function in surgeons poses significant methodological challenges. Only recently have investigators attempted to explore this area (Leff et al., 2006; Wanzel et al., 2007). Studies evaluating cortical activation in surgeons have relied upon correlations between surgical tasks and abstract visuo-spatial tests more suited to the confines of an fMRI environment (Wanzel et al., 2007). The application of fNIRS would facilitate analyses of brain function during surgical tasks rather than surrogates, providing valuable information regarding expertise development.

The purpose of this paper is to evaluate the impact of experience on a task, on serial cortical haemodynamic changes detected using fNIRS. To demonstrate this effect we evaluate cortical function in expert and novice surgeons during serial evaluations of a surgical knot-tying task. This task is taught to junior surgical trainees at an early stage in their career and is practised frequently in many common general surgical procedures. For a beginner, the task is not trivial. Surgical knot-tying requires the learner to integrate visuomotor sequences into smooth coordinated spatio-temporal arrangements, and to our knowledge represents the most complex task evaluated by fNIRS to date. Surgical novices are anticipated to progress from a highly attention demanding initial phase where they learn to hold the tie, place the throws and move their hands to an integrative stage where performance is more fluid (Reznick and MacRae, 2006). Since knot-tying is executed frequently in many surgical procedures, expert surgeons are less likely to be ruminating the sub-steps, enabling allocation of resources to other more crucial aspects of the operation.

In this study, 62 subjects performed a surgical knot-tying task, comprising five cycles of task repetitions. Prefrontal cortical activation was recorded during each cycle of the task using fNIRS.

We hypothesised that variations in cortical activation patterns would be associated with acquisition of the surgical task. Specifically, we hypothesised that prefrontal activation would attenuate in the context of performance improvements in novices. Minimal fluctuation in cortical activation is anticipated in trained subjects (registrars and consultants).

Materials and methods

Subjects

Sixty-two, right-handed healthy male subjects participated in this study, following Local Regional Ethics Committee (LREC) approval. Written informed consent was obtained from each participant prior to enrolment. Participants included 19 consultant surgeons (mean age $\pm$ S.D. = 46.3 ± 7.6), 21 surgical registrars (mean age $\pm$ S.D. = 31.3 ± 4.1 years) and 22 medical students (mean age $\pm$ S.D. = 21.9 ± 1.9 years), all of whom were recruited from Imperial College London, United Kingdom (UK).

Consultants and registrars were recruited from various surgical disciplines, including general surgery (11 consultants and 15 registrars), cardiothoracics (2 consultants and 2 registrars), vascular surgery (3 consultants and 1 registrar), urology (2 consultants and 3 registrars) and gynaecology (1 consultant). Open surgical knot-tying is generic to operative procedures performed within these specialties. Medical students were only recruited providing they had no prior experience of open surgical knot-tying, enabling them to be classified as surgical 'novices'. All subjects were strongly right hand dominant (Edinburgh Handedness Inventory; Oldfield, 1971). Participants had neither a previous history nor family history of neuropsychiatric disorders. Subjects were requested to refrain from drinking alcohol for 24 h prior to the investigation.

Training

Medical students were trained in a single one-hour session to perform hand tied surgical reef knots, as taught on the Basic Surgical Skills (BSS) course, using a standard bench top knot-tying trainer (Ethicon Ltd, New Jersey, USA). One to one intensive training involved a surgeon trainer demonstrating hand tied surgical reef knots, in the following standardised manner. First, four throws of a surgical reef knot were demonstrated without commentary, followed by an identical demonstration with commentary. The steps involved in formulating knots were then segregated into 12 subsets (three per throw) and again these were demonstrated with and without commentary. Participants were then required to perform each subset four times. A period of 15 min was allocated for limited practice on the task with feedback provided from the trainer where necessary. Finally, at the end of the brief practice session, each subject had to successfully form four throws of a hand tied surgical reef knot using suture material (2/0 Polysorb) in order to proceed to serial functional neuroimaging evaluation. All medical students successfully completed this task and no subjects were excluded from the study at this stage.

Task paradigm

The motor task under investigation consisted of four throws of a hand tied surgical reef knot performed using 2/0 Polysorb on a bench top knot-tying trainer (Ethicon Ltd, New Jersey, USA). The task was introduced in a block design manner. Blocks consisted of

baseline rest (30 s), trial (self paced surgical reef knot) and post-trial rest periods (20–30 s). The rest periods prior to the first and after the last trial were 30 s duration. All other inter-trial rest periods lasted 20 s. During rest periods, the subjects sat on a chair in front of the desk, with their eyes focused solely on the knot-tying trainer and their hands placed on the table at motor rest. Following a verbal prompt (“start”) from the investigator, subjects formulated four throws of a surgical reef knot, self paced on the knot-tying trainer from the resting position. As soon as the last throw of the knot had been laid, subjects returned to the resting position (“rest”).

Behavioural performance

Participants were monitored using the Imperial College Surgical Assessment Device (ICSAD) (Taffinder et al., 1999).

This system consists of a commercially available electromagnetic field generating device (Isotrak II, Polhemus, United States) and two non-contact sensors that are attached to the dorsum of the subject's hands in a standardised manner. Cartesian x , y and z coordinates were converted into dexterity data such as, the time taken and the number of movements required to complete the task as well as the distance or ‘pathlength’ travelled by each hand. Data was acquired at 20 Hz and a Gaussian filter was applied to reduce noise. All trials were videotaped and time-linked to motion analysis data for retrospective viewing to ensure the knot-tying technique correctly matched that performed during training.

Functional near infrared spectroscopy (fNIRS)

Optical measurements were obtained using the ETG-4000 Optical Topography System (Hitachi Medical Co., Japan) using a

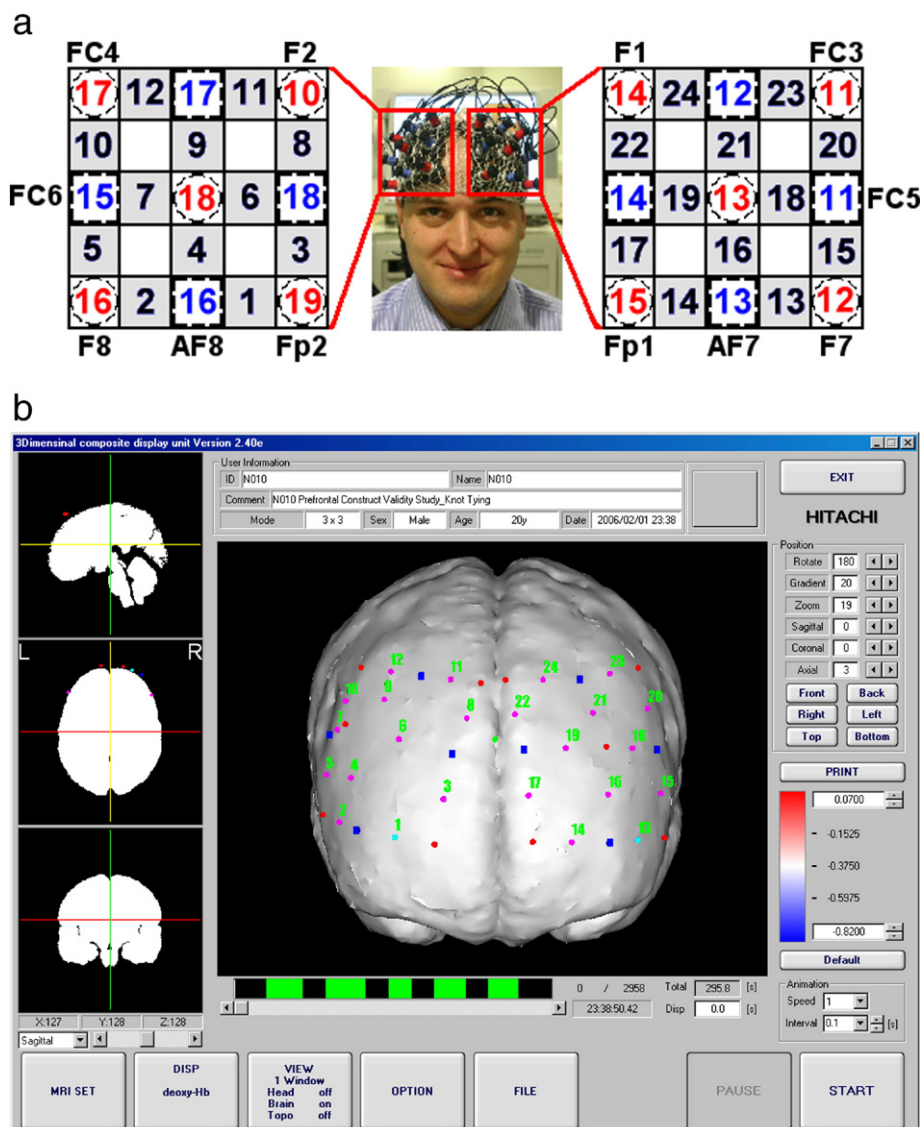


Fig. 1. Panels (a–b): The experimental design, illustrating (a) the positions of NIR light sources (red), detectors (blue) and locations of corresponding channels numerically labelled in shaded boxes. One 3×3 cm array was centered on the right prefrontal cortex and a similar array was centered on the left prefrontal cortex. The approximate International 10–20 and 10–10 system cranial locations of each optode array are provided. (b) Optode location obtained by transferring topographic data to 3D cortical surface image of a maximum probability MRI atlas. The figure illustrates the approximate location of NIR light sources (solid red circles), detectors (solid blue circles) and channels (solid purple circles/green numbers) on the cortical surface.

24-channel array of optodes. A 3×3 cm probe configuration was employed, consisting of 10 optical fibre sources (semiconductor laser diodes), emitting NIR light at 695 nm and 830 nm and eight optical fibre receivers coupled to avalanche photodiode detectors, each of which detected the reflected light of its neighbouring and surrounding emitters. Each laser diode was modulated at a unique frequency for each wavelength and channel. Following detection, signals passed through a switching circuit set according to the configuration of the measuring probes, and were resolved by 48 lock-in amplifiers locked to the modulation frequency of the light source. Optical signals were sampled at a rate of 10 Hz, digitised and transformed according to their wavelength and location resulting in relative changes in light intensities at 0.1 s intervals for each of the 24 measuring positions.

A schematic illustration of optode placement is shown in Fig. 1 (a). Optodes were secured in standard thermoplastic holders (inter-optode distance 30 mm) and fastened to the participant's head using a flexible plastic helmet, affixed with surgical bandage (Surgifix, Colorline, Italy). The optodes were placed bilaterally over the prefrontal cortices and positioned according to the International 10–20 system of electrode placement (Jasper, 1958). Concerning the right hemisphere, optodes were positioned such that the lowermost medial emitter was centered on Fp2, the lowermost lateral on F8, and the central emitter on F4. Optode positioning on the left hemisphere symmetrical mirrored that of the right. Specifically, the lowermost medial emitter was centered on Fp1, the lowermost lateral on F7 and the central emitter on F3. According to recent evidence (Okamoto et al., 2004a,b), the probabilistic anatomical correlations for these cranial markers are the granular area 9, frontopolar area 10, prefrontal area 11, triangular area 45 and orbital area 47 (Brodmann, 1909), which cytoarchitecturally correspond to the PFC (Crosby et al., 1962).

The position on the participant's head contacting the 10 emitting fibres and eight receiving fibres as well as four base points (nasion, inion, and bilateral external auditory meati) were measured using an electromagnetic probe positioning digitiser. To better appreciate the locations of each channel, the position of the probes from a representative case was transferred to an MRI image using a 3-dimensional Composite Display Unit (Hitachi Medical Co., Japan). Probe positions were projected on to the cortical surface of a 3D maximum probability atlas (Hammers et al., 2003), to determine

cortical structures underlying each measuring position as illustrated in Fig. 1(b).

Data analysis

Optical data were subjected to an identical pre-processing procedure using the functional Optical Signal Analysis (fOSA) program (Koh et al., 2006), applying the Modified Beer–Lambert Law to convert relative changes in light intensities to oxyhaemoglobin (HbO₂), deoxyhaemoglobin (HHb) and their sum total haemoglobin (HbT). Data was baseline corrected, decimated to 1 Hz, and detrended to remove system drift and unrelated physiological signals.

For a given subject and channel, averaged 'baseline rest' haemoglobin values were calculated by averaging 5 s of data, prior to the start of each knot-tying stimulus. Similarly, averaged 'trial' haemoglobin data was calculated for each stimulus period, by averaging 10 s of data, commencing 3 s after the onset of knot-tying. This stimulus window was selected on the basis that the averaged Time to Peak (TTP) for both HbO₂ and HHb has been reported to be within 10–13 s of a motor stimulus (Maki et al., 1995; Watanabe et al., 1996; Obrig et al., 2000; Boas et al., 2001; Miyai et al., 2001; Jasdzewski et al., 2003; Strangman et al., 2003; Huppert et al., 2006). Averaged 'trial' values were subtracted from averaged 'baseline rest' values to obtain a single value for both Δ HbO₂ and Δ HHb for every knot-tying episode. The Δ HbO₂ and Δ HHb values were statistically analysed in order to determine the location of channels demonstrating significant variations in cortical haemodynamics across knot-tying trials.

The Statistical Package for Social Science (SPSS v14, Chicago, Illinois) was used to statistically analyse both functional neuroimaging and behavioural data. Optical and performance data were non-normally distributed, could not be centralised using common transformations and were therefore analysed using non-parametric tests of significance. The Friedman Test was used to determine if statistically significant fluctuations had occurred in technical performance and cortical haemodynamic data across knot-tying trials. A value of $p \leq 0.05$ was assumed to be statistically significant. The Wilcoxon Sign Rank Test for related non-parametric data was used to determine which knot-tying interval (i.e. 1–2, 2–3, etc.) was responsible for the greatest change in

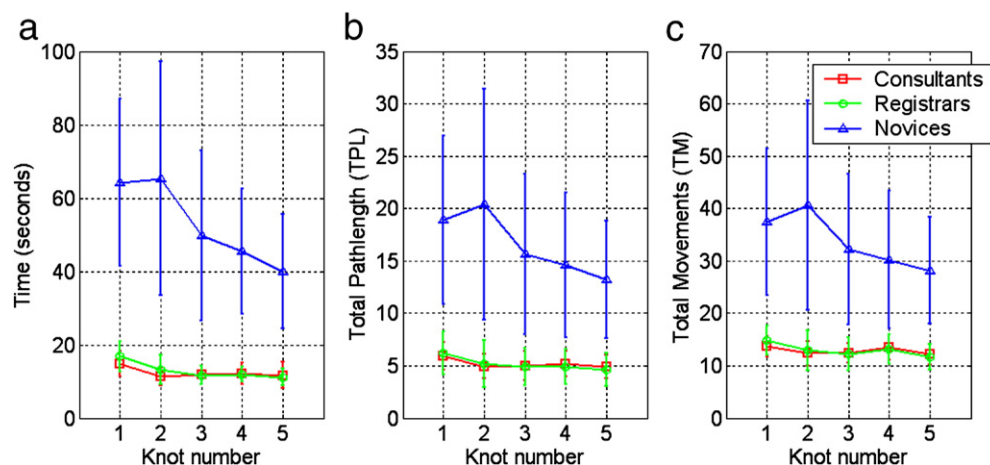


Fig. 2. Plots illustrating the variation in technical performance across knot-tying trials according to the level of surgical experience. Subplots correspond to (a) Time, (b) Total pathlength (TPL) and (c) Total movements (TM), to complete knot-tying trials.

performance and fluctuations in cortical haemodynamics. A Bonferroni corrected value of $p \leq 0.01$ ($0.05/4$) was assumed to be statistically significant.

Results

Technical performance

Fig. 2 illustrates variations in technical performance across five knot-tying trials, according to surgical experience. As anticipated, better technical performance was observed in trained surgeons compared to medical students. The greatest improvements in technical performance observed in novices, occurred between the second and third trials. Minimal fluctuations were observed in knot-tying performance amongst trained surgeons, aside from an anticipated improvement between the first and second knots (it is easier to formulate a second surgical knot upon a pre-laid knot than it is to formulate the first knot). Statistical analysis of performance data revealed significant fluctuations ($p \leq 0.01$) for each dexterity metric in all three experience groups. In trained surgeons, significant improvements ($p \leq 0.01$) in performance were only observed between the first and second knot-tying sessions. In contrast, maximal improvement in medical students occurred between the second and third knot tying trials. Medical students were significantly faster ($p \leq 0.01$), and required fewer movements ($p \leq 0.01$) to complete the third versus the second trial. Shorter pathlengths were required during the third trial but this parameter did not reach statistical significance ($p = 0.036$).

Task-related changes of cortical haemodynamics

Figs. 3(a–c) summarises the trend in ΔHbO_2 and ΔHHb across five knot-tying trials as a function of surgical experience. Cortical haemodynamic change varied considerably across trials in medical students. In contrast, comparatively minimal fluctuations in haemodynamic change were observed across trials in trained surgeons. Striking similarities in haemodynamic changes were identified across prefrontal channels in medical students. Specifically, an increase in ΔHbO_2 and a decrease in ΔHHb occurred between the first and second trials in nearly all 24 prefrontal channels. A decrease in ΔHbO_2 and an increase in ΔHHb were also observed toward the end of experiment, between the third and fourth, and fourth and fifth trials. To emphasise these transitions in cortical haemodynamic change, sample time course traces for a typical medical student subject are provided in Fig. 4. In trained surgeons, no readily identifiable patterns of haemodynamic change occurred across prefrontal channels.

The results of statistical analysis are shown in Tables 1 and 2. Table 1 illustrates the locations of channels in which significant fluctuations in ΔHbO_2 and ΔHHb occurred across trials for each experience group. Greater fluctuations in cortical haemodynamics occurred in medical students compared to trained surgeons. Table 2 illustrates the results of the post-hoc analysis (Wilcoxon Sign Rank Test), for each experience group, in which both ΔHbO_2 and ΔHHb were compared across knot-tying trial intervals (i.e. 1–2, 2–3, etc.) to determine the transition of maximal change. Similar patterns of cortical change were identified in prefrontal channels in medical students. Statistically significant ($p \leq 0.01$) increases in ΔHbO_2 coupled to decreases ΔHHb were commonly observed between the first and second trials. In addition, statistically significant ($p \leq 0.01$)

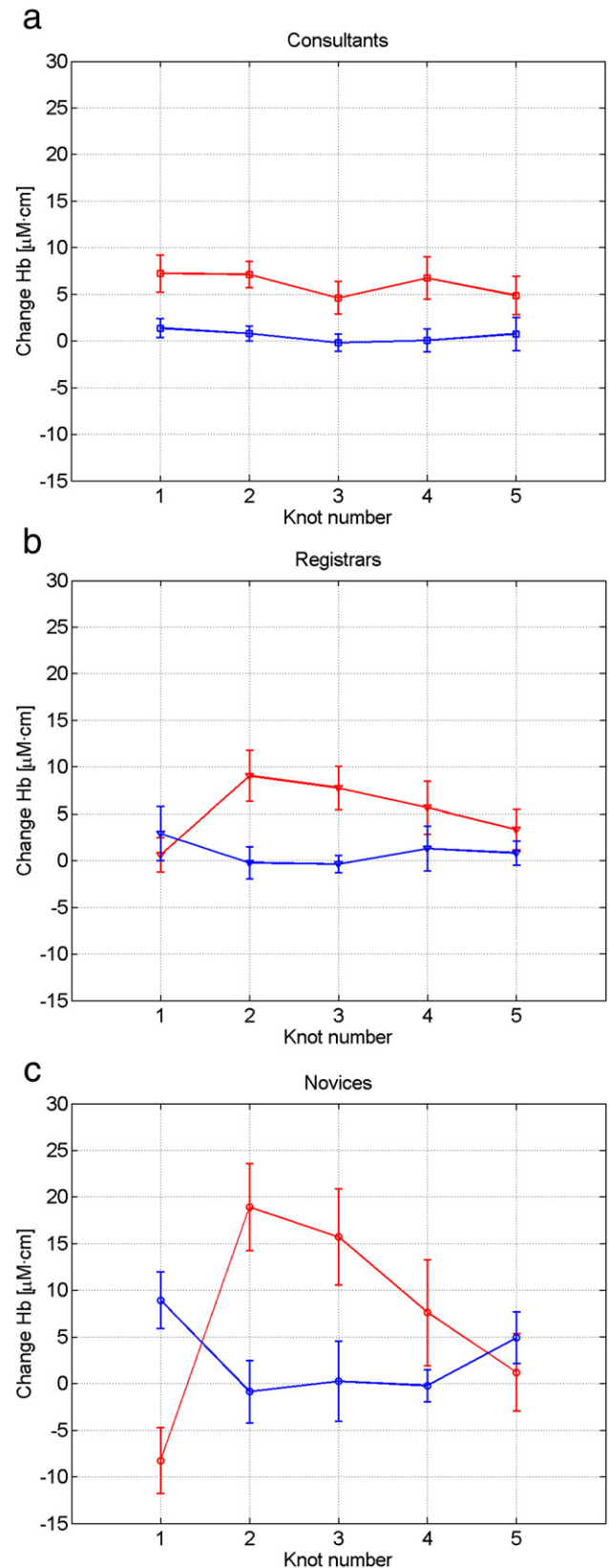


Fig. 3. Panels (a–c): Illustrates variations in mean ΔHbO_2 (red line) and ΔHHb (blue line) averaged across all 24 prefrontal channels for five knot-tying trials according to surgical experience. Subplots correspond to (a) Consultants, (b) Registrars and (c) Novices. Error bars represent 95% CI of the mean.

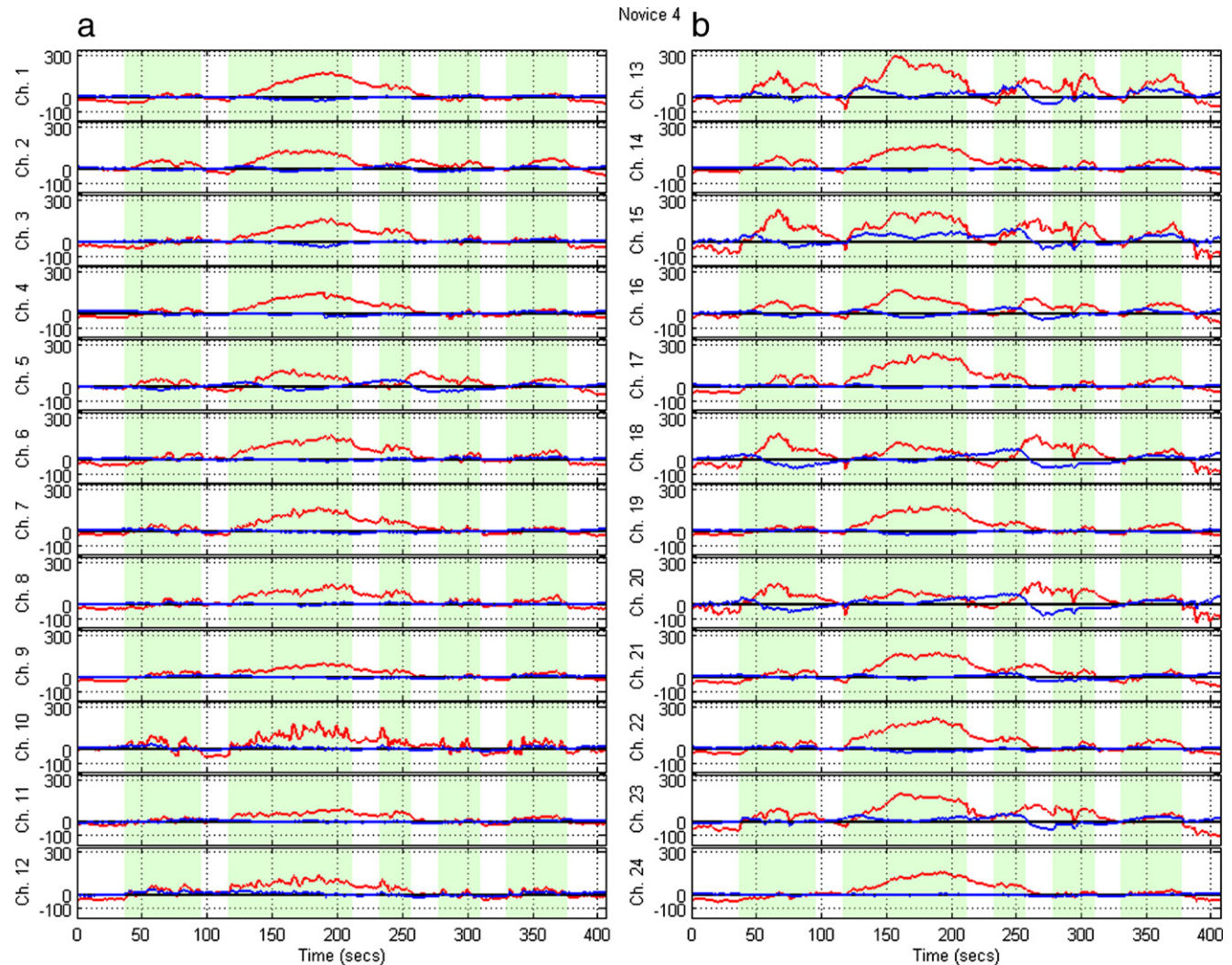


Fig. 4. Panels (a–b): The typical time course haemodynamic changes for 24 prefrontal cortical channels (ch) in a typical novice subject. Traces illustrate the changes in HbO₂ (red line) and HHb (blue line). Variable stimulus knot-tying episodes (light green shaded vertical windows) are interspersed with fixed rest periods (white vertical windows). Units of haemodynamic change are given as $\mu\text{M cm}$.

decreases in ΔHbO_2 and increases in ΔHHb were commonly identified between the third and fourth, and fourth and fifth trials respectively. No consistent patterns of haemoglobin change were identified in trained surgeons.

Discussion

We have demonstrated that task-related increases in HbO₂ and decreases in HHb dynamically change with acquisition of a surgical task. Patterns of haemodynamic change appear to be dependent on years of experience on the task. In this example, a significant transition in prefrontal excitation occurred between the first and second knot-tying trials in surgical novices. This transition was associated with improvements in technical performance during subsequent trials. Considerable improvements in technical performance by the end of the experiment were associated with reduced demands for prefrontal activation in these subjects. In contrast, minimal fluctuations in prefrontal haemodynamic changes were

associated with more stable technical performance in trained surgeons.

Role of the prefrontal cortex in motor skill learning

The PFC has been shown to play an important role in novel task performance when motor performance is unrefined and attention demands are high (Lacourse et al., 2005). Enhanced PFC activation in the initial trials in surgical novices may be explained by ‘attention to action’, since intense concentration is known to lead to greater prefrontal activation compared to unattended performance (Rowe et al., 2002). The ability to hold, manipulate and monitor information online is referred to as visuospatial working memory (VSWM) (Baddeley, 1986), and evidence from neuroimaging studies in healthy subjects (Owen et al., 1996; Courtney et al., 1997; Slotnick and Moo, 2006) suggests that the dorsolateral prefrontal cortex (DLPFC) subserves working memory. It is conceivable that in the initial trials surgical novices had to retain

Table 1
The results of Friedman analysis for ΔHbO_2 and ΔHHb across knot-tying trials for each experience group

Haemoglobin parameter	Consultants		Registrars		Novices	
	ΔHbO_2	ΔHHb	ΔHbO_2	ΔHHb	ΔHbO_2	ΔHHb
Channel	–	–	–	–	–	–
1	–	–	–	0.011	0.003	0.000
2	0.029	–	–	–	–	–
3	–	–	–	–	0.019	0.000
4	0.017	–	–	0.015	0.002	0.000
5	–	–	0.026	–	–	0.016
6	–	–	–	–	0.05	0.000
7	–	–	–	–	0.015	0.000
8	–	–	–	–	0.000	0.000
9	–	–	–	–	0.001	0.000
10	0.023	–	–	–	–	–
11	–	–	–	–	0.000	0.000
12	–	–	–	–	0.001	0.010
13	–	0.049	–	–	–	–
14	0.041	–	–	0.002	0.003	0.000
15	–	0.032	–	0.016	–	–
16	–	–	–	–	0.004	0.001
17	–	–	–	0.002	0.001	0.001
18	–	0.027	–	–	–	0.000
19	–	0.034	–	0.022	0.008	0.000
20	–	–	0.046	–	–	0.024
21	–	–	–	–	0.002	0.000
22	–	–	–	–	0.003	0.000
23	–	–	–	–	0.011	0.008
24	–	0.034	–	0.021	0.005	0.000

ΔHbO_2 =change in oxyhaemoglobin, ΔHHb =change in deoxyhaemoglobin. Data are significant p values ($p \leq 0.05$) to 3 decimal places.

visuospatial images of knot-tying technique using a rehearsal system, maintain this information in the delay between practice and assessment, and subsequently select the correct responses.

Enhanced PFC and ACC activations during erroneous responses suggests these regions are crucial for error detection and online monitoring of performance (Carter et al., 1998; Sarter et al., 2006). Having repeatedly observed refined ‘expert’ performance during training sessions, it is suspected that novices registered poor co-ordination, unnecessary movements and breaks in flow as technical errors in performance. Error detection was likely to be most prominent during early knot-tying trials in novices possibly accounting for PFC activation. The ventrolateral prefrontal cortex (VLPFC) is thought to direct attention to relevant stimuli in memory, silence irrelevant aspects of the same stimulus

(Kostopoulos and Petrides, 2003), and processes spatial cues in novel performance enabling the learner to discover new spatio-temporal relationships between the wrists and hands (Puttemans et al., 2005). In spite of brief offline practice it is suspected that novices were still discovering new spatiotemporal arrangements during the initial trials.

Interestingly, global PFC responses observed in the current study cannot be extrapolated to all scenarios in which motor performance is novel. Comparatively minimal PFC activation has been observed as subjects learn to perform simple sequential finger movements (Lacourse et al., 2005) and bimanual in-phase/out-phase wrist movements (Debaere et al., 2004). Some of the disparity between these findings may be explained by differences in motor paradigms, neuroimaging techniques and training schedules. For example, all movement conditions employed by Debaere et al. (2004) were practised intensely over two days, possibly minimising the magnitude of prefrontal responses. In contrast, novices in the current investigation were only briefly exposed to the task, ensuring unrefined performance throughout initial trials. Additionally, surgical knot-tying is considered to be more cognitively challenging than either sequential finger movements or bimanual wrist movements. Complex goal orientated behaviours modulate the degree of prefrontal activation. Tasks such as surgical knot-tying that require complex cognitive strategies; multiple sequential command processing and conscious manipulation of stored information are likely to result in profound PFC activations.

In the current study, attenuation in PFC activation associated with practice-related performance improvements was observed toward the end of the experiment in surgical novices. These findings are supported by practice dependent prefrontal plasticity associated with bimanual co-ordination (Debaere et al., 2004), visuomotor association (Iacoboni et al., 1996), cyclical wrist movements (Floyer-Lea and Matthews, 2004; Puttemans et al., 2005) and non-surgical knot-tying (Tracy et al., 2003), all of which require the chaining together of components of learned sequences into new spatio-temporal arrangements in a similar manner to surgical knot-tying. Despite different methodologies and paradigms, these studies demonstrate that prefrontal and premotor cortices are engaged in early phases of motor learning whereas primary and supplementary motor regions are preferentially recruited when the task is well-learned. Regions activated early in practice such as the PFC and ACC are thought to act as a ‘scaffold’ to support novel demands, during unrefined effortful motor performance (Petersen et al., 1998). It is possible that practice-related ‘pruning’ of functional activations occurred as

Table 2
Results of the post-hoc analysis (Wilcoxon Sign Rank) of ΔHbO_2 and ΔHHb across knot-tying intervals in consultant surgeons, registrars and medical students

Group	Channels demonstrating $p \leq 0.01$							
	ΔHbO_2				ΔHHb			
Knot-tying interval	1–2	2–3	3–4	4–5	1–2	2–3	3–4	4–5
Consultants		2(↓), 10(↓)	2(↑)		24(↑)			
Registrars					4(↓), 14(↓), 16(↓)	17(↓), 24(↑)	17(↑)	
Medical students	4(↑), 6(↑), 8(↑), 9(↑), 11(↑), 12(↑), 16(↑), 17(↑), 21(↑), 22(↑), 24(↑)		1(↓), 3(↓), 4(↓), 6(↓), 8(↓), 9(↓), 11(↓), 12(↓), 14(↓), 17(↓), 19(↓), 21(↓), 22(↓), 24(↓)		1(↓), 3(↓), 4(↓), 6(↓), 8(↓), 9(↓), 11(↓), 18(↓), 19(↓), 20(↓), 21(↓), 22(↓), 24(↓)		8(↑)	3(↑), 4(↑), 5(↑), 7(↑), 14(↑), 18(↑), 20(↑), 22(↑), 23(↑)

Data are channels (direction of change) demonstrating significant change ($p \leq 0.01$) between one trial and the next. Direction of change (↑) or (↓) are highlighted.

novices developed enhanced technical dexterity (Petersen et al., 1998). The observed variation in PFC and ACC activation may signal a change in cognitive strategy across knot-tying trials. These regions are thought to work in tandem to monitor action value, process response outcome and modify behaviour (Walton et al., 2007). PFC and ACC activation may signify desired versus actual response comparison, enabling corrective behaviours that facilitate improved performance in latter trials.

Practice dependent cortical plasticity is thought to be highly dependent on the task domain. Surgical knot-tying represents a complex paradigm with both sequential motor and cognitive components, making more detailed interpretations of the current analysis more challenging. Practice-related changes associated with simple motor tasks comprise a ‘redistribution’ effect resulting from upregulation within the primary motor cortex (M1) (Karni et al., 1998) and activation decreases in generic control regions such as the PFC and ACC (Kelly and Garavan, 2005). To confirm whether the attenuation in PFC activation towards the end of the experiment is associated with increases in M1 activation suggestive of a kinematic memory trace would require more extensive cortical coverage of fNIRS optodes. It is possible that PFC redundancy reflects ‘reorganisation’ of activation foci indicative of a shift in cortical processing towards the end of practice (Kelly and Garavan, 2005). This is consistent with findings from other studies of practice-related plasticity, in which shifts from effortful to automated control have been observed for cognitive tasks (Petersson et al., 1999).

In summary, we have demonstrated that prefrontal excitation evoked by five trials of surgical knot-tying differs in surgical novices from trained surgeons. Specifically, practice dependent variations in prefrontal activation in novices are as anticipated and are associated with improvements in technical performance. Interestingly, a similar study of non-surgical knot-tying has demonstrated persistent PFC activation in subjects with difficulty learning manual tasks (Tracy et al., 2003). Other studies involving visual memory (Buchel et al., 1999) visuomotor (Haier et al., 1992) and cognitive abilities (Beauchamp et al., 2003) have observed more efficient brain function in subjects better at performing these tasks. The relationship between better performance, reduced PFC activation and neuronal efficiency is fascinating and has obvious implications for the selection and assessment of future surgeons. Future work should aim at evaluating the effects of intense practice dispersed over protracted durations. More extensive cortical coverage of optodes would enable functional ‘redistribution’ of activation foci to be measured. This would allow us to test the hypothesis that a shift in cortical processing accompanies the transition from novice to expert surgeon.

Finally, cortical haemodynamics (especially HbO₂) can be influenced by variations in heart rate and blood pressure occurring during cognitively demanding tasks (Tachtsidis et al., in press). Theoretically, fluctuations in systemic haemodynamics may have contributed to the results. However, we anticipate that the greatest contribution of systemic haemodynamics would have occurred during the first knot-tying block. This effect should have resulted in a larger HbO₂ signal during the first knot-tying trial versus the second; the exact opposite of the pattern observed. This phenomenon is not thought to reflect a delayed time to peak in cortical haemodynamics; rather we anticipate that poor performance in the first knot-tying block leads to enhanced attention and concentration in the second block. For studies involving surgical drills, concomitant fNIRS and systemic monitoring is not trivial,

since many devices for continuous non-invasive heart rate and blood pressure recording require cuffs to be placed around the digits. We are exploring the use of continuous wireless heart rate monitoring using a custom built ear piece, which should enable us to quantify the contribution of the systemic effect.

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